Operating Systems

Chapter 4: Process Scheduling | Year 2 lecture handout

4.1 The scheduling problem

A computer with one processor can run only one process at a time. Every other ready process waits in a queue. The scheduler is the part of the operating system that decides which process runs next and for how long. Good scheduling keeps the processor busy, keeps response time short, and treats every process fairly.

4.2 When the scheduler runs

The scheduler is called when a process moves from running to waiting, when a process moves from running to ready, when a process moves from waiting to ready, and when a process terminates. The first and the last of these are non preemptive: the running process gives up the processor on its own. The other two are preemptive: the operating system takes the processor away.

4.3 Scheduling criteria

Throughput is the number of processes completed per unit of time. Turnaround time is the time from submission to completion. Waiting time is the total time spent in the ready queue. Response time is the time from submission to the first response. No algorithm optimises all four at once, so every scheduler is a compromise.

4.4 First come first served

First come first served runs processes in the order they arrive. It is simple and it never starves a process, but a long job at the front of the queue makes every short job behind it wait. This is the convoy effect.

4.5 Shortest job first

Shortest job first runs the process with the smallest next CPU burst. It gives the smallest average waiting time when every burst is known in advance. In practice the next burst is only an estimate, taken from the length of the previous bursts.

4.6 Round robin

Round robin gives every ready process a turn in a fixed time slice, called a quantum. When the slice ends the process goes to the back of the ready queue and the next process runs. The size of the slice decides how the system feels.

A time slice that is too short turns the processor into a switchboard: too much time goes to saving and restoring context, and too little to useful work. A time slice that is too long makes round robin behave like first come first served, and a student clicking a button waits behind a batch job. A slice of about ten to one hundred milliseconds is the usual compromise.

Worked example. Five processes arrive at the same time with bursts of 8, 4, 2, 1 and 1 milliseconds. With a quantum of 2 milliseconds the first process finishes in its fourth turn, so response time stays short for the four short jobs. With a quantum of 8 milliseconds the short jobs wait for the whole first burst.

4.7 Priority scheduling

Priority scheduling always runs the ready process with the highest priority. It can starve a low priority process forever. Ageing fixes this by raising the priority of a waiting process a little on every pass, so no process waits without end.

4.8 Summary

The scheduler chooses the next process. Preemption lets the system take the processor back. Throughput, turnaround, waiting and response time pull in different directions. Round robin protects response time, shortest job first protects average waiting time, and priority scheduling needs ageing to stay fair.